VaahanSetu

UPDATED SOFTWARE REQUIREMENTS SPECIFICATION

Version 2.0 | Updated passenger input and multi-stop route model

Course: CS1207 Object Oriented Programming Lab
Group No. 6 VaahanSetu
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Revision Summary

Version 2.0 updates only the passenger-demand input method and route representation. The core purpose, administrator and crowd management officer roles, Java-based desktop planning system, bus eligibility rules, allocation comparison, reporting, simulation, security and optional heatmap remain unchanged except where they must support passenger-provided journey details and intermediate stops.

- Passenger input: passengers register their intended journey using a mobile interface by selecting a route, boarding stop and destination stop. Boarding and alighting are confirmed in the system so estimated on-board counts can be updated.
- Multi-stop routes: every route contains an ordered list of stops. Demand and crowding are calculated for each segment between consecutive stops, not only for the route as a whole.

1. Introduction

1.1 Purpose

This SRS defines the requirements for VaahanSetu Version 2.0. It describes users, inputs, system behavior, constraints and outputs so the team can implement and test a common scope. This revision introduces passenger-submitted journey details and stop-by-stop crowd estimation while retaining the original bus allocation and comparison workflow.

1.2 Product Scope

VaahanSetu is a Java-based desktop planning application for bus allocation and crowd management. An administrator selects buses and routes for a planning session. Each route contains an ordered sequence of stops. Passengers use a mobile-friendly interface to register a journey by choosing the route, boarding stop and destination stop. The system uses these records, together with boarding/alighting confirmations, to estimate passenger load on each route segment.

The crowd management officer reviews the collected demand and initiates allocation. The system checks bus capacity, schedules, journey duration, availability and rest periods, then proposes bus-to-route assignments. It compares the proposed allocation with the original scheduled allocation using the same demand data and reports accommodated passengers, unmet demand and unused capacity. Demand may also be simulated every two minutes, after which allocation and comparison results are recalculated.

The system is a planning aid and demonstration, not a live transport authority system. Mobile journey registrations and confirmations are user-provided data; location assistance may help but shall not be treated as infallible proof of boarding or alighting.

1.3 Definitions

Term	Definition
Route	A journey path represented by an ordered sequence of stops.
Stop	A designated boarding, alighting or intermediate point on a route.
Route segment	The part of a route between two consecutive stops.
Journey registration	A passenger's mobile submission of route, boarding stop and destination stop.
Boarding confirmation	A passenger confirmation that they have boarded the bus.
Alighting confirmation	A passenger confirmation that they have left the bus.
Estimated on-board load	The number of registered passengers estimated to be on the bus or route segment at a given time.
Demand	Expected passenger need, derived from passenger journey registrations and any clearly labelled planning adjustments.
Bus capacity	Maximum passenger count the bus is allowed to carry, as stored in the bus record.
Unmet demand	Passengers who cannot be accommodated by the capacity assigned to the relevant route segment.
Unused capacity	Allocated passenger capacity not needed to accommodate the estimated demand.
Rest period	A scheduled interval during which a bus must not be assigned to a route.

1.4 References and Assumptions

Requirements and scope decisions are supplied by the VaahanSetu team. No external transport data source is assumed. The team supplies sample buses, routes, ordered stops, schedules, capacities, rest periods and user accounts. Passengers have access to a mobile phone and can submit journey details, but the system must account for missing, duplicate, cancelled or unconfirmed registrations. Mobile input alone does not guarantee that a passenger actually boards a bus.

2. Overall Description

2.1 Product Overview

The system has two main interfaces: a Java desktop interface for the administrator and crowd management officer, and a passenger-facing mobile-friendly interface for journey registration and boarding/alighting confirmation. The exact mobile technology is not prescribed by this SRS. The system stores bus and route data, collects passenger journey details, calculates stop-by-stop estimated load, proposes allocations and displays comparison results.

2.2 Main Workflow

1. Administrator logs in and selects buses and multi-stop routes for the planning session.

2. Passengers submit route, boarding stop and destination stop through the mobile interface.

3. Passengers confirm boarding and, when leaving, alighting; the system updates estimated on-board and segment loads.

4. The crowd management officer reviews the stop-wise demand and submits the planning data.

5. The system validates inputs and checks bus schedules, capacity, availability, route duration and rest periods.

6. The system calculates a proposed allocation and evaluates the original scheduled allocation using the same demand snapshot.

7. The system displays accommodated passengers, unmet demand and unused capacity for both scenarios, including segment-level crowd information.

8. If simulation is started, demand is adjusted every two minutes, then the allocation and comparison are recalculated.

2.3 Constraints

- The core planning application shall be a Java desktop application. The passenger interface may be a mobile app or mobile-friendly web interface; its technology is not fixed by this SRS.
- The system shall use a database or equivalent persistent store for buses, routes, ordered stops, schedules, capacities, rest periods and login records. Passenger journey records must be retained at least for the active planning session and its calculations.
- A route shall contain an ordered sequence of at least two stops. Boarding stop must occur before destination stop in the route sequence.
- Passenger load shall be calculated per segment between consecutive stops. A passenger contributes to the load only on segments from their boarding stop up to, but not including, their destination stop.
- A bus shall not be assigned to overlapping journeys and must be available for the full journey and outside its rest period.
- Estimated passenger load shall not exceed the bus's permitted capacity in a valid proposed assignment. If demand exceeds available capacity, the system shall report the unmet demand.
- The demand simulation shall update every two minutes, changing demand by 5–10 passengers on multiple randomly selected routes/segments while preventing negative demand. Simulated values must be labelled as simulated.
- The allocation algorithm is not prescribed, but it must obey the defined constraints and aim to reduce unmet demand.

2.4 Assumptions and Dependencies

- Passengers provide correct route, boarding stop and destination details and confirm boarding/alighting when prompted.
- If a passenger fails to confirm boarding or alighting, the system shall identify the record as unconfirmed or stale according to a team-defined rule; it shall not silently claim exact live occupancy.
- Duplicate registrations, cancelled journeys and invalid stop sequences must be detected or flagged.
- Bus schedules, route duration, ordered stops and rest periods are available in stored records.
- Original-versus-proposed comparisons use the same demand snapshot and stop sequence.
- Allocation results need not be retained after the planning session ends, but persistent master data remains stored.
- A geographic heatmap is optional and depends on suitable stop coordinates or route-path data.

3. Specific Requirements

3.1 Functional Requirements

F1: The system shall provide login for the administrator and crowd management officer and identify each user's role after successful authentication.

F2: The system shall display stored bus records available for the planning session.

F3: The system shall display route records, including each route's ordered list of stops.

F4: The administrator shall be able to select buses and multi-stop routes for a planning session.

F5: The system shall make the administrator's selected buses and routes visible to the crowd management officer.

F6: The passenger interface shall allow a passenger to select a route, a boarding stop and a destination stop, then submit the journey registration.

F7: The system shall validate that the selected boarding and destination stops belong to the selected route and that the boarding stop occurs before the destination stop.

F8: The system shall record boarding and alighting confirmations and update estimated on-board counts and loads for the affected route segments.

F9: The system shall flag duplicate, cancelled, incomplete or unconfirmed journey records and shall not treat unconfirmed mobile registrations as guaranteed actual occupancy.

F10: The system shall calculate passenger demand separately for every segment between consecutive stops. A registered passenger contributes to segments from the boarding stop to the stop immediately before the destination stop.

F11: The system shall allow the crowd management officer to review stop-wise and segment-wise demand before allocation.

F12: The system shall determine bus eligibility using schedule, route duration, availability, capacity and scheduled rest periods.

F13: The system shall not assign a bus if the journey conflicts with another assignment, extends into a required rest period, or the bus is otherwise unavailable for the full journey.

F14: The system shall ensure that a bus is assigned to at most one route at a time within the proposed allocation.

F15: The system shall generate and display proposed bus-to-route assignments intended to reduce unmet demand while respecting all defined constraints, including segment-level capacity.

F16: The system shall compare the original scheduled allocation and proposed allocation using the same validated demand snapshot and route-stop data.

F17: For each scenario, the system shall display accommodated passengers, unmet demand and unused bus capacity. It shall also display estimated crowd/load by route segment.

F18: After the officer starts simulation, the system shall update demand every two minutes by increasing or decreasing demand by 5–10 passengers on multiple randomly selected routes or segments, without allowing negative demand.

F19: After each simulated update, the system shall recalculate proposed assignments, segment loads and comparison results, and clearly label simulated values.

F20: The system may display a route heatmap or schematic using segment crowd levels; the core application shall work without this optional visualization.

3.2 Use Case Model — Actors and Use Cases

Actors: Administrator, Crowd Management Officer and Passenger. Main use cases: Log in; View buses and multi-stop routes; Select buses and routes; Register passenger journey; Confirm boarding/alighting; Review segment demand; Calculate allocation; Compare allocations; Run demand simulation; View revised results.

3.3 Use Case Specifications

U1 — Log In

Actors: Administrator and Crowd Management Officer. Preconditions: application is running and the user has a registered account. Basic flow: user enters credentials; system validates them and identifies the role; system opens the permitted interface. Invalid credentials produce an error and allow retry. If the login store is unavailable, the system reports that authentication cannot be completed.

U2 — Select Buses and Multi-stop Routes

Actor: Administrator. Preconditions: administrator is logged in and records exist. Basic flow: view bus records; view routes and their ordered stops; select buses and routes; make the selection visible to the officer. The administrator may change the selection before demand is finalized. If records are unavailable, the system displays an empty list and blocks meaningful allocation.

U3 — Register Passenger Journey

Actor: Passenger. Basic flow: open the mobile interface; select route; select boarding stop and destination stop; review and submit the journey. The system validates route membership and stop order, then records the journey. If a stop is invalid, the system asks the passenger to correct it. The interface should collect only information needed for the journey and should not expose personal passenger details to other passengers.

U4 — Confirm Boarding and Alighting

Actor: Passenger. Basic flow: passenger confirms boarding when getting on the bus and confirms alighting at the destination. The system updates estimated on-board count and the affected segment loads. Optional location assistance may support the workflow, but location alone shall not be represented as guaranteed proof. If confirmation is missing or contradictory, the record is flagged for review or handled by a defined timeout rule.

U5 — Calculate and Compare Allocation

Actor: Crowd Management Officer; system performs calculation. Preconditions: selected buses/routes and a validated demand snapshot are available. The system computes segment demand, checks schedule, capacity, availability and rest periods, creates a proposed allocation, evaluates the original schedule with the same demand snapshot, and displays accommodated passengers, unmet demand, unused capacity and segment-level crowd. If capacity is insufficient, remaining unmet demand is shown. Buses with invalid schedule data are excluded and the issue is reported.

U6 — Run Demand Simulation and Refresh Results

Actor: Crowd Management Officer. Preconditions: an initial allocation exists and the officer starts simulation. Every two minutes, the system randomly changes demand by 5–10 passengers on multiple selected routes/segments, prevents negative values, recalculates allocation and comparison statistics, and displays revised results. The cycle continues while active. If recalculation fails, an error is displayed and failed output is not presented as valid.

4. Other Non-functional Requirements

4.1 Performance

- NFR-P1: For the team's sample dataset, the system should display the initial allocation and comparison results promptly after valid demand is available.
- NFR-P2: Simulation updates shall follow the defined two-minute interval while active.
- NFR-P3: Revised assignments, segment loads and comparison statistics should be displayed before the next update interval.
- NFR-P4: The desktop interface should remain responsive while displaying records and results.
- NFR-P5: Passenger journey submission and confirmation should update the relevant estimated segment loads without requiring the entire session to be restarted.

4.2 Safety, Security and Data Quality

- NFR-S1: Role-specific functions shall require successful login where applicable.
- NFR-S2: Bus/route selection shall be restricted to the administrator; allocation review and simulation controls shall be restricted to the officer; passengers may submit and manage only their own journey records.
- NFR-S3: Demand values and route-stop sequences shall be validated; demand and simulated load shall not become negative.
- NFR-S4: Simulated or estimated passenger loads shall be clearly distinguished from verified live counts.
- NFR-S5: Missing or invalid schedule data shall not be silently treated as proof of bus availability.
- NFR-S6: Passenger data shall be limited to what is necessary for journey registration and crowd estimation, protected from unauthorized access, and not displayed as personally identifiable information on public screens.
- NFR-S7: The system shall handle duplicate, cancelled, incomplete and unconfirmed journey records in a consistent, documented way.

5. Other Requirements

A route heatmap is optional. If implemented, it may show the route and its intermediate stops, using colors to represent estimated crowding relative to allocated capacity on each segment. Geographic display requires suitable stop coordinates or route-path data. Without that data, the core application shall still work.

The system is intended for one planning session at a time. Master data such as buses, routes, stops, schedules, capacities, rest periods and login records shall persist. Passenger journey records and allocation results may be retained only for the active session unless the team chooses to add a retention policy.

Appendix A — Key Data and Calculation Rules

A. Route and segment model

Example route: Stop A → Stop B → Stop C → Stop D. The route has three segments: A–B, B–C and C–D. A passenger travelling from B to D contributes to B–C and C–D, but not A–B. A passenger travelling from A to C contributes to A–B and B–C, but not C–D.

B. Basic crowd calculations

For each segment, estimated load is the number of valid, active passenger journeys whose boarding stop is at or before the segment's start and whose destination stop is after the segment's start in the route order, adjusted for recorded boarding/alighting confirmations under the team's chosen data rules.

For a segment with demand D and allocated capacity C: accommodated passengers = min(D, C); unmet demand = max(0, D – C); unused capacity = max(0, C – D). These formulas assume a consistent passenger-counting rule and must not double-count a bus's capacity across simultaneous overlapping journeys.

C. Implementation decisions to finalize

- Choose how passengers confirm boarding/alighting (for example, in-app confirmation or QR scan). GPS/geofence assistance is optional and should not be the sole proof of occupancy.
- Define when an unconfirmed registration expires, how cancellations are handled, and how duplicate journey submissions are detected.
- Define how simulation affects demand while real passenger submissions continue, so simulated and user-submitted values remain distinguishable.
- Define the allocation algorithm and how capacity is checked on every route segment.
- Define data retention and privacy handling for passenger journey records.